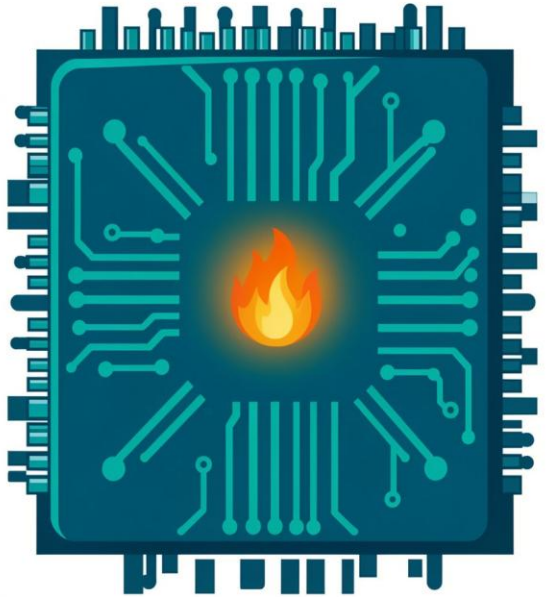
Beyond FLOPs

An Empirical Study of Computational Complexity and Actual Energy Consumption in Deep Learning Inference



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Discoverize — Research Paper Presentation

The Problem: AI Has an Energy Cost

- Deep learning models are becoming increasingly computationally demanding.
- Models are commonly evaluated using accuracy, speed, FLOPs, and parameter count.
- These metrics describe complexity and performance, but do not directly represent measured energy consumption.

Can model complexity metrics tell us how much energy a model actually consumes?

From AI Performance to Green AI

Conventional model evaluation: accuracy + speed + complexity.

Green AI: accuracy + performance + resource efficiency.

Green AI encourages consideration of computational cost and resource efficiency alongside model performance.

Focus of this study: INFERENCE ENERGY.

Research Gap

01

Model complexity : FLOPs and parameter count.

02

Execution.

03

Hardware + runtime behaviour.

04

Measured energy consumption.

Complexity metrics do not necessarily capture the full energy behaviour observed during execution.

WHAT WE INVESTIGATE

Research Question

01 — COMPLEXITY
FLOPs + Parameter Count

02 — ENERGY
Measured CPU Package Energy

03 — PERFORMANCE
Inference Time

To what extent do FLOPs and parameter count reflect actual energy consumption during deep learning inference?

Controlled experiment → 4 models → 5 trials/model → 500 inferences/trial

Experimental Methodology

01	02	03
4 pretrained models	Same CPU environment	224 × 224 input
04	**05**	**06**
20 warm-up inference runs	5 independent trials	500 inference runs per trial

07
Measure + analyze

Measurements: FLOPs, parameter count, inference time, CPU package energy.

Analysis: mean, standard deviation, Pearson correlation.

Energy refers to CPU package energy measured during the inference workload, not whole-laptop electricity consumption.

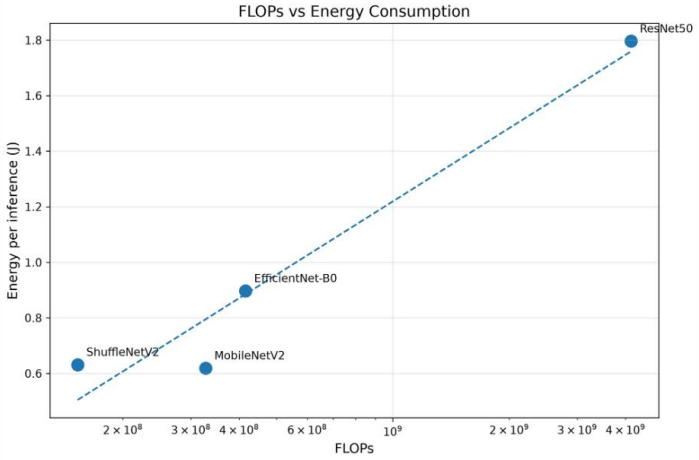
Models & Experimental Setup

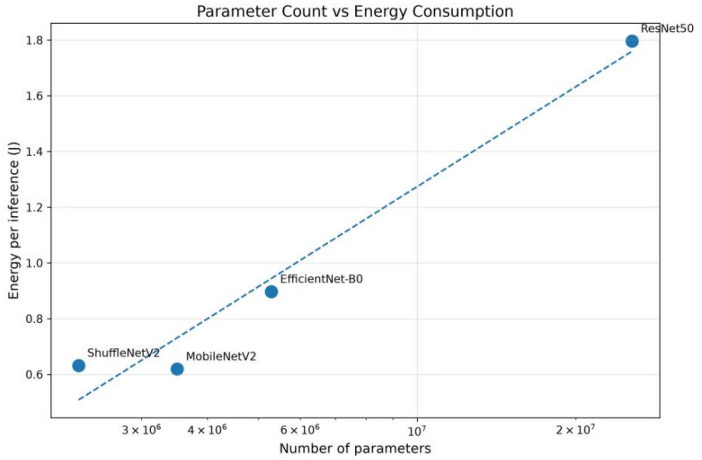
MODEL	PARAMETERS	FLOPS
ShuffleNetV2	2,278,604	152,708,792
MobileNetV2	3,504,872	327,486,720
EfficientNet-B0	5,288,548	415,145,024
ResNet50	25,557,032	4,133,742,592

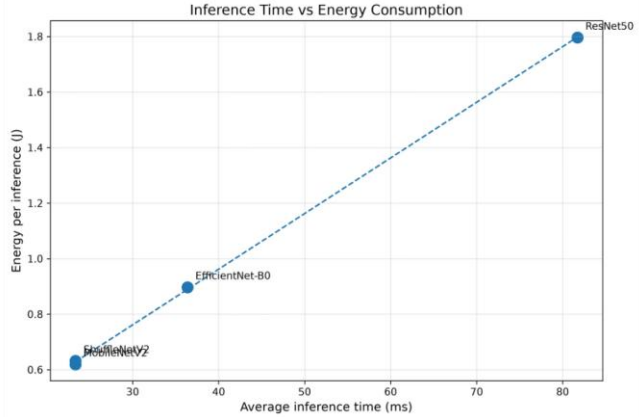
CPU: Intel Core Ultra 7 155H · CPU-only · Input: 1 × 3 × 224 × 224 · Warm-up: 20 runs · Trials: 5 valid trials per model · Inference runs: 500 per trial

The FLOPs values use the THOP profiling convention used in the experiment.

Results: What Does the Data Show?







FLOPs ↔ Energy:
r = 0.981

Parameters ↔ Energy:
r = 0.990

Inference Time ↔ Energy:
r= 0.9999

Strong overall relationships were observed between complexity metrics, inference time, and measured energy..

The Key Finding: Lower FLOPs ≠ Lower Energy

SHUFFLENETV2

152.7M FLOPS

0.632 J

Inference time · 23.35 ms

>2× FLOPs

MOBILENETV2

327.5M FLOPS

0.619 J

Inference time · 23.33 ms

MobileNetV2 required more than twice the FLOPs of ShuffleNetV2, yet consumed slightly less measured energy per inference.

FLOPs are useful indicators of overall computational behaviour, but they are not complete proxies for measured energy consumption.

Conclusion: Measure the Cost, Not Just the Complexity

01 — WHAT WE FOUND
FLOPs and parameter count showed strong overall associations with measured inference energy

02 — WHAT IT MEANS
Complexity metrics are useful — but they are not complete proxies for actual energy consumption

03 — WHAT COMES NEXT
More model architectures
Larger model samples
CPU vs GPU comparisons
Different hardware platforms
Different workloads and deployment environments

.

If we want to build more sustainable AI systems, model evaluation should consider not only what a model can do, but also what it costs to run.

Future work: more model architectures · larger model sample · different hardware platforms · GPU vs CPU comparison · different workloads and deployment environments